# WhatsApp AI Bot Setup Guide

MakeMyStore website (Next.js on Vercel) + Meta WhatsApp Cloud API + Google Gemini Flash-Lite

**Purpose of this document.** A record of everything done to connect the WhatsApp test number to the same AI brain as the website chat bot: files added, Meta settings, Vercel variables, what worked, what failed and why, and what is still left. **It contains no tokens or secrets.** Never commit real keys to Git.

## 1. Final result (status at the time of writing)

| Item | Status |
|---|---|
| Website AI chat bot (Gemini) | Already working before this work. Not changed in behaviour. |
| WhatsApp bot on Meta TEST number (+1 555 667 7803) | **WORKING.** Message from the owner phone gets an AI reply. |
| Permanent access token (System User) | Created and added to Vercel. Bot still replies. |
| Real business number (client-facing) | **NOT DONE YET.** Test number cannot be used publicly or in ads. |
| Click-to-WhatsApp ads | **NOT DONE YET.** Needs a real registered number linked to the Facebook Page. |
| Lead saving / email alerts / human handover | **NOT BUILT.** Optional next features. |

## 2. How it works (architecture)

```
1. Customer sends a WhatsApp message to the business number
2. Meta WhatsApp Cloud API sends a webhook POST to:
      https://www.makemystore.online/api/whatsapp
3. The route checks the Meta signature (App Secret)
4. It loads that customer's recent messages and the shared prompt
5. It calls Gemini Flash-Lite and gets the answer
6. It sends the answer back with the Access Token (Graph API)
7. Customer receives the reply on WhatsApp
```

- **Webhook GET**: Meta calls it once to verify the callback URL using the Verify Token.
- **Webhook POST**: Meta calls it for every customer message. The code always answers 200 so Meta does not retry.
- The bot only answers text messages. For photos, voice notes etc. it asks the customer to type.
- Replies are free-form 'service' messages: allowed within 24 hours after the customer's last message.

## 3. Code files added to the project

Three files were produced. In the file names the double underscore marks the folder path; the final name inside the project is shown below.

| File | Action | Path in project | What it does |
|---|---|---|---|
| 1_NEW__lib_chatbot__system-prompt.ts | ADD | lib/chatbot/system-prompt.ts | Shared 'brain': builds the system prompt from lib/data.ts and persona.ts. Used by both bots so answers always match. |
| 2_REPLACE__app_api_chat__route.ts | REPLACE | app/api/chat/route.ts | Website bot. Same behaviour as before; it now imports the shared prompt instead of holding its own copy. |
| 3_NEW__app_api_whatsapp__route.ts | ADD | app/api/whatsapp/route.ts | WhatsApp webhook (GET verify + POST messages). New folder app/api/whatsapp/ must be created. |

**Why every file is called route.ts:** in Next.js the FOLDER decides the URL. app/api/chat/route.ts = /api/chat (website), app/api/whatsapp/route.ts = /api/whatsapp (WhatsApp). They are different files in different folders. Never save the WhatsApp file into the chat folder or it overwrites the website bot.

### Behaviour built into the WhatsApp route

- Checks the `x-hub-signature-256` header with the App Secret (rejects fake requests with 401).
- Keeps the last 10 messages per customer phone number (in memory, resets on cold start).
- Limit: 30 messages per customer per hour, so nobody can run up the Gemini bill.
- Ignores duplicate deliveries (same message id) and empty or very long messages (cut at 1000 characters).
- Adds WhatsApp rules to the prompt: plain text only, reply in the customer's language, no form (asks name and need in chat).
- Strips the website-only marker [[COLLECT_CONTACT]] so customers never see it.
- Model: DEFAULT_GEMINI_MODEL from lib/gemini.ts (gemini-3.5-flash-lite). Override with GEMINI_MODEL if Google renames it.

## 4. Environment variables (Vercel, Production)

Add in Vercel: Project > Settings > Environment Variables. **Redeploy after every change**, otherwise the running site keeps the old values.

| Name (exact spelling) | What it is | Where to get it |
|---|---|---|
| GEMINI_API_KEY | Paid Google AI key (already existed for the website bot) | Google AI Studio |
| WHATSAPP_VERIFY_TOKEN | A short text YOU invent (example: makemystore_secret_123). Must match what you type in Meta webhook setup. **Not** a Meta token. | You choose it |
| WHATSAPP_PHONE_NUMBER_ID | ID of the business number the bot replies from (about 15 digits). Not secret. | Meta app > Use cases > Customize > Step 1 / Step 2 |
| WHATSAPP_ACCESS_TOKEN | Very long token starting with EAA. Lets the bot SEND replies. Temporary one lasts about 24 hours; System User token can be permanent. | Step 1 'Generate token' (temporary) or Business Settings > System users (permanent) |
| WHATSAPP_APP_SECRET | Short secret (about 32 characters). Used to verify Meta's signature. | Meta app > App settings > Basic > App secret > Show |
| GEMINI_MODEL (optional) | Overrides the default model name. | Google AI Studio |

**Common mistake we made:** pasting the very long access token into WHATSAPP_VERIFY_TOKEN. The long token belongs ONLY in WHATSAPP_ACCESS_TOKEN. Missing WHATSAPP_APP_SECRET caused every real message to fail with 401.

# 5. Step-by-step: what was done in Meta and Vercel

## A. Put the code online

- Add the 3 files, push to GitHub, Vercel deploys. Wait until the deployment says Ready.
- Check: opening https://www.makemystore.online/api/whatsapp in a browser shows 'Forbidden'. That is correct (it proves the route exists). A 404 means the file is not deployed.

## B. Create the Meta app

- developers.facebook.com > My Apps > Create App > type Business.
- On 'Add use cases' tick ONLY 'Connect with customers through WhatsApp', then Next. Choose the business portfolio.

## C. Test number and token (Use cases > Customize > Step 1. Try it out)

- Meta provides a free test number (+1 555 667 7803) with a Phone Number ID and a WhatsApp Business Account (WABA) ID.
- Click Generate token for a temporary access token. Add your own phone as a recipient and confirm the code. Send the 'hello world' message to check.
- Copy the App Secret from App settings > Basic.

## D. Vercel variables

- Add the 4 WhatsApp variables from section 4, then Redeploy.

## E. Webhook (Step 2. Production setup > Configure Webhooks)

- Callback URL: `https://www.makemystore.online/api/whatsapp` (use the www address, the same host Vercel serves).
- Verify token: the same text as WHATSAPP_VERIFY_TOKEN. Click Verify and save. Vercel logs show GET 200 when it works.
- In the Webhook fields list make sure **messages** is Subscribed (Meta turned it on automatically).

## F. Publish the app

- In App settings > Basic set Privacy Policy URL (https://www.makemystore.online/privacy). Then Publish (status must show Published/Live).
- While Unpublished, Meta only sends fake dashboard test messages, not real phone messages.

## G. Subscribe the WhatsApp account to the app (the fix that made real messages arrive)

- Open developers.facebook.com/tools/explorer. Meta App = Business. Add permissions whatsapp_business_management and whatsapp_business_messaging. Click Generate Access Token.
- Set method to POST, URL box to `WABA_ID/subscribed_apps`, click Submit. Result must be `{"success": true}`.
- Check with GET on the same URL: the app 'Business' should be listed.

### H. Permanent token (business.facebook.com > Business Settings > Users > System users)

- Add a NEW system user (example: whatsapp-bot), role Admin. Do not reuse the existing 'Conversions API System User' (that is for the Pixel).
- Add assets: the app 'Business' with Full control, and the WhatsApp account with Full control.
- Generate token for the app, expiry Never, tick whatsapp_business_messaging and whatsapp_business_management. Copy it once into WHATSAPP_ACCESS_TOKEN and Redeploy.

### I. Test

- From the recipient phone send a message to the test number. Vercel Logs should show POST /api/whatsapp with 200 and the phone gets the AI reply.

## 6. What worked and what did not (problem log)

Every problem we hit, what the symptom looked like, the real cause and the fix. Use this when something breaks again.

| Symptom | Cause | Fix |
|---|---|---|
| Meta 'Verify and save' failed, or logs showed GET 403 from user agent facebookexternalua | Verify token typed in Meta did not match WHATSAPP_VERIFY_TOKEN (or box was blank/masked when re-clicking Verify). | Retype the token exactly in Meta. Do not keep re-clicking Verify. GET 200 in logs = verified. |
| Dashboard 'Send to server' test returned POST 401 | WHATSAPP_APP_SECRET was missing. The long access token had been pasted into WHATSAPP_VERIFY_TOKEN and the other 3 WhatsApp variables did not exist. | Reset verify token to a short text; add ACCESS_TOKEN, PHONE_NUMBER_ID, APP_SECRET; Redeploy. |
| Vercel log: 'WhatsApp send error: 400 ... #131030' | Meta's sample test message comes from a fake phone number, so WhatsApp refuses the reply (recipient not in allowed list). | Expected and harmless. POST 200 means the webhook and signature are correct. Test with a real phone. |
| Real phone message got no reply and no POST appeared in logs | (1) App was Unpublished, so only dashboard test messages are delivered. (2) The test WhatsApp account was not subscribed to the app. | Publish the app (needs Privacy Policy URL) AND run POST WABA_ID/subscribed_apps in Graph API Explorer. After both, real messages arrived. |
| Graph Explorer was left on method DELETE with the default URL | DELETE on subscribed_apps would remove the link between the account and the app. | Always check the method before Submit. Re-run POST WABA_ID/subscribed_apps to restore ({"success": true}). |
| Possible redirect problem | Callback URL without www while Vercel serves www.makemystore.online. | Use https://www.makemystore.online/api/whatsapp in Meta. |
| Reply stops after about a day | Temporary access token (24 hours) expired. Messages still arrive but sending fails. | Use the System User permanent token (step H) or generate a new temporary one and Redeploy. |
| Registering a real number: page froze after entering the phone number | Browser or extension problem or a temporary Meta glitch. | Wait, refresh, try Incognito/another browser, or use business.facebook.com > WhatsApp Manager > Phone numbers > Add phone number. |
| Vercel shows SUPABASE_SERVICE_ROLE_KEY 'Needs Attention' | Unrelated to WhatsApp (contact form / Supabase). | Check later, it does not block the bot. |

### What works with the TEST number

- Phones that you added as recipients in Step 1 can chat with the bot and get Gemini replies.
- Dashboard test messages reach the webhook (POST 200).
- Replies in the customer's language (English / Urdu / Arabic) because the prompt tells Gemini to mirror the customer.

### What does NOT work with the TEST number

- Public customers cannot message it. Only listed recipient phones work.
- It cannot be used as the CTA number in ads or on the website button.
- Its temporary token expires in about 24 hours (use the permanent System User token).
- It is not your brand: no business name, no real phone number.

## 7. What is left to do (go live)

| # | Task | How |
|---|---|---|
| 1 | Register the REAL business number | Step 2. Production setup > Register your WhatsApp phone number. Enter number WITHOUT the leading 0 (example: 7462 230676 with UK +44). Confirm the SMS/call code. Set display name and category. |
| 2 | Switch the bot to the real number | Copy the NEW Phone Number ID into WHATSAPP_PHONE_NUMBER_ID. Add the NEW WhatsApp account to the system user with Full control. Run POST NEW_WABA_ID/subscribed_apps in Graph Explorer. Redeploy. |
| 3 | Payment method | Step 2 > Add payment (needed for business-initiated template messages). |
| 4 | Business verification | Step 3. Optional but recommended (2-10 business days). Needs company documents. Raises limits and shows business name. |
| 5 | Click-to-WhatsApp ads | Link the real number to the Facebook Page. In Ads Manager choose Messaging and WhatsApp as destination. |
| 6 | Website WhatsApp button | Link format: https://wa.me/COUNTRYCODE+NUMBER?text=Hi, I found you on your website (no + and no leading 0). The pre-filled text tells you the source. |
| 7 | Optional features | Save leads to Supabase + email alert, tag source (ad / website / direct), 'talk to a human' handover, persistent chat memory in Supabase. |

**About the +44 number that already runs WhatsApp Business:** a number used in the normal WhatsApp or WhatsApp Business app normally has to be deleted from the app before it can be registered on the Cloud API, which loses its chats. Meta also has a mode where the Business app and the API share one number; it was not confirmed for this UK number. In Meta's registration screen, look for a 'connect your WhatsApp Business app' (QR code) option before choosing the SMS-code option. Once a number is on the Cloud API (without that mode) you read chats in Meta WhatsApp Manager, not in the phone app. If chats matter, use a fresh SIM instead.

## 8. Costs (verify on Meta's pricing page, it changes)

- **Replies to customers who message you first (service messages): free.** This is all the bot does today.
- Conversations started from Click-to-WhatsApp ads get an extra free window. Guides say 72 hours; Meta announced 7 days from 21 September 2026.
- **You messaging first** (marketing, utility, authentication templates) is charged per message and varies by country. The bot does not do this yet.
- Gemini Flash-Lite: pay per use through your Google paid tier. Watch Google AI Studio > Usage.
- Ads budget: paid to Meta.

- Vercel Hobby is free but is meant for personal, non-commercial use. Check Vercel's terms before real client traffic; Pro may be required.
- SIM / number: normal SIM cost only. Business verification itself is free.

## 9. Security checklist

- Never commit .env.local or any token to Git. Keep secrets only in Vercel Environment Variables.
- Access tokens appeared in some screenshots during setup. Regenerate: click Generate Access Token again in Graph Explorer, and create a fresh System User token, then update Vercel and Redeploy.
- Do not share the App Secret: anyone with it can send fake messages to your webhook.
- Keep the signature check in route.ts. Do not remove it.
- Rate limit (30 per hour per phone) protects your Gemini bill. Keep it.

## 10. Quick troubleshooting checklist

| Check | Where |
|---|---|
| Is the latest Vercel deployment Ready after the last variable change? | Vercel > Deployments |
| Does https://www.makemystore.online/api/whatsapp show 'Forbidden' in a browser? | Browser. 404 = file not deployed |
| Webhook callback URL (with www), verify token, and 'messages' = Subscribed | Meta > Step 2 > Configure Webhooks |
| App status is Published / Live | Meta > Publish |
| Account subscribed to app: GET WABA_ID/subscribed_apps lists the app | Graph API Explorer |
| Sender phone is a listed recipient (test number only) | Meta > Step 1 > Send a message |
| New POST /api/whatsapp line appears when a message is sent. Status 200 = good, 401 = App Secret, 500 = code or variable | Vercel > Logs (set filter to Last 30 minutes, click refresh) |
| POST 200 but no reply and a red error in Messages column: expired or wrong access token / wrong Phone Number ID | Regenerate token, update variable, Redeploy |

## 11. Useful links

- Meta for Developers: developers.facebook.com
- Graph API Explorer: developers.facebook.com/tools/explorer
- Business Settings / System users: business.facebook.com
- WhatsApp Manager: business.facebook.com (WhatsApp Manager)
- Google AI Studio (keys and usage): aistudio.google.com
- Vercel project: vercel.com (Logs, Deployments, Environment Variables)